Where Does Entropy Collapse Go? Measuring the Visitation Channel That Token-Level Entropy Control Misses

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Abstract. Reinforcement learning with verifiable rewards (RLVR) drives a language model's entropy toward zero, and the standard remedies act token by token. We ask where the lost entropy goes. The change in trajectory entropy splits exactly into a *local* channel, sharpening at the states the policy visits, and a *visitation* channel, a change in which states it visits; token-level control acts only on the first. Measured between checkpoints of a 1.5B model, 0.90 of the entropy GRPO loses by step 100 goes through the visitation channel, and a registered replication on two further seeds gives 0.93 and 0.91. In the run we followed, most of the loss occurs within the first 50 steps. We show why the local channel cannot be extended to cover the other, and derive the visitation term as a per-token advantage that sums to zero over sibling rollouts, so aggregate entropy cannot register it. A five-arm experiment over three paired seeds bears this out: changing the damping mechanism moves aggregate entropy, changing only the visitation signal barely does. We make no accuracy claim; the paper is an account of where entropy collapse happens and of what aggregate entropy curves cannot show.

Keywords: Large language models · Reinforcement learning · Entropy collapse · Training diagnostics · Reasoning

1 Introduction

Policy-gradient fine-tuning with verifiable rewards sharpens a language model's output distribution. Measured as the mean per-token entropy, the sharpening is fast, monotone and, past a point, coincident with the end of useful learning. The standard remedies spend part of the update budget slowing it down: an entropy bonus, a KL anchor, a clip on the tokens that lose most entropy, or a per-token re-weighting of the policy loss by a first-order estimate of the entropy each token's update destroys [5]. All of them act on the conditional distribution at a state the policy has reached. This paper asks a prior question: when RLVR loses entropy, how much of the loss is sharpening at the states the policy visits, and how much is a change in which states it visits?

The distinction matters because token-level control can see only the first, and the omission is structural. Fig. 1 shows the case. At a state with two equiprobable continuations the conditional distribution is as entropic as it can be, and its

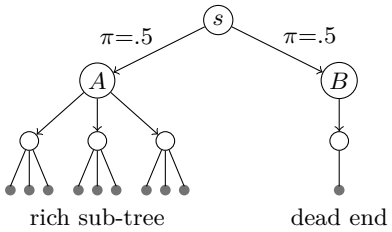


Fig. 1. At s the conditional entropy is maximal and its first-order change is 0 (local channel), yet moving mass from A to B removes A's downstream diversity (visitation channel). Token-level entropy control sees only the first.

first-order entropy change under a small push is zero. Yet shifting mass from a continuation that leads into a rich sub-tree to one that leads into a dead end removes all the downstream diversity the first carried. The branch is locally safe; which branch is visited is not.

Writing $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution over whole responses, its gradient splits, by the product rule and with no approximation, into a *local* term that holds visited states fixed — what a per-token estimate measures — and a *visitation* term that differentiates the probability of reaching each state. The local estimate vanishes at a flat branch point, correctly, since the entropy loss there is second order; the visitation term's prefactor is largest at those same points.

How much of real collapse goes through each channel is an empirical question, and the same split answers it exactly between two checkpoints. From the base model to step 100 of GRPO, 0.90 of the lost trajectory entropy goes through visitation, and the share replicates on two other seeds (Section 4). Most of what is called entropy collapse is therefore not sharpening at fixed states but the policy settling on paths along which the base model was already confident — the part token-level control does not touch. Within the scope of what we measured, this has three practical consequences. First, finer per-token weights are the wrong lever: the two damping variants we tested leave the split where GRPO puts it, so preserving diversity calls for objectives that act on which paths are sampled and reinforced, such as outcome-level exploration [13] or rollouts that diversify prefixes [2]. Second, timing matters: by step 50 the run we followed had already lost -590.4 of the -616.0 nats it lost by step 100, so an intervention switched on once entropy looks low arrives after most of the collapse. Third, mean token entropy is the wrong monitor: it cannot register a change in which paths are visited (Section 5), whereas the split needs only samples and forward passes at two checkpoints, so it can serve as a data-driven diagnostic for any LLM training run, including released checkpoints.

Our contributions are:

– **A measurement of where collapse goes**: an exact between-checkpoint split of the entropy that training loses, which assigns 0.90 of GRPO's loss

to the visitation channel, replicated on two further seeds, and usable as a diagnostic on any two checkpoints (Section 4).
- **Why token-level control cannot reach it**: the local estimator vanishes at flat branch points, where the visitation prefactor is largest (Section 3).
- **An estimable visitation term and its diagnostic consequence**: a per-token advantage A_H that sums to zero over siblings, so aggregate token entropy cannot diagnose it (Section 5).
- **A test of that consequence**: five arms that separate the damping mechanism from the visitation signal (Section 5.1).

2 Background and Related Work

A prompt is x, a response $y = y_{1:T}$, and $s_t = (x, y_{<t})$ the state before token t. GRPO [12] samples n responses per prompt and broadcasts the group-standardized reward to every token.

Entropy collapse and its control. Entropy decay under policy-gradient fine-tuning of reasoning models is tied to the stall of improvement [4] and is countered by an asymmetric clip [19], by restricting updates on high-covariance tokens [4], by KL control with reference resets [10], or by selective entropy regularization, since a naive bonus tends to explode [7]. Training on a minority of high-entropy "forking" tokens suffices [15], and several schemes shape advantages or rewards with token entropy [3,14,20]. STEER [5] leaves the advantage untouched and multiplies each token's clipped surrogate by a weight derived from the first-order change in the conditional entropy at a *fixed* state after one step,

$$\Omega(s) \propto -\mathbb{E}_a\Big[\tfrac{\mathbb{I}_{\text{clip}} A}{\pi_{\text{old}}}\,\pi_\theta(a)\big(1-\pi_\theta(a)\big)\big(\log \pi_\theta(a) + H\big)\Big], \tag{1}$$

evaluated at the sampled token and mapped into a weight $\alpha_u \in [\lambda_{\min}, 1]$ that decreases exponentially in $|\Omega_u|$ relative to its batch maximum, so the weight can only attenuate. All of these act on conditional distributions at visited states, that is, on the local channel.

Trajectory-level diversity. Token-level entropy can stay high while the diversity of final answers falls [16], and outcome-level exploration targets that diversity directly [13]. The visitation channel is a mechanism for the first observation. Tree-structured rollouts give intermediate decisions their own reward credit [6,18,2] and choose where to branch [2,8]; we use the same structure to supply a baseline for remaining entropy rather than for reward. The identity we rest on is the entropy term of the maximum-entropy RL objective, whose policy gradient is standard there [11,9]; what we add is its use to measure real collapse and to locate token-level control.

Table 1. The two channels are oppositely ordered around flat branch points. The top token has mass q, the rest is spread uniformly over m tokens; P(split) is the probability that n=8 samples do not all agree.

distribution	H	P(split)	$\mathbb{E}\lvert\Omega\rvert/\lvert A\rvert$	$\mathbb{E}[1-\pi_a]$
q=.99	0.10	0.08	0.092	0.020
q=.9, tail	1.25	0.57	**1.129**	0.190
q=.6, m=3	1.11	0.98	0.457	0.587
uniform, k=10	2.30	**1.00**	**0.000**	**0.900**
uniform, k=100	4.61	**1.00**	**0.000**	**0.990**

3 Two Channels

The autoregressive factorization gives $\mathcal{H}(\pi_\theta \mid x) = \mathbb{E}_{y\sim\pi_\theta}[\sum_t H(\pi_\theta(\cdot \mid s_t))]$. Both the integrand and the measure depend on θ; with $f_\theta(y) := \sum_t H(\pi_\theta(\cdot \mid s_t(y)))$, the product rule and the score-function identity give, with no approximation,

$$\nabla_\theta \mathcal{H} = \underbrace{\mathbb{E}\big[\nabla_\theta f_\theta(y)\big]}_{\text{(L) local}} + \underbrace{\mathbb{E}\big[f_\theta(y)\,\nabla_\theta \log \pi_\theta(y \mid x)\big]}_{\text{(V) visitation}}. \tag{2}$$

(L) holds the visited states fixed and is what Ω estimates. (V) holds per-state entropies fixed and asks how the probability of reaching each state changes. The two can have opposite signs, so driving (L) to zero does not bound $\nabla_\theta \mathcal{H}$.

The local channel cannot be repaired. One might expect the channels to coincide, since branch points are uncertain positions. Three facts rule this out. (i) $\Omega \propto \pi_a(1-\pi_a)(H - I_a)$ with $I_a = -\log \pi_\theta(a \mid s)$ and $H = \mathbb{E}[I_a]$, so $|\Omega|$ is set by the spread of surprisal, not the level of entropy. (ii) At a uniform branch point $I_a = H$ for every a, so $\Omega \equiv 0$. (iii) That zero is correct: the logit derivative of the conditional entropy, $-\pi_\theta(a)(\log \pi_\theta(a) + H)$, vanishes at a uniform distribution, so the local loss is entirely second order. The visitation prefactor, in contrast, is $1 - \pi_a$, which tends to 1 as the distribution flattens — exactly where a group of samples is certain to diverge. The local signal vanishes where the visitation prefactor is largest, so no refinement of the local term recovers what it omits. Table 1 evaluates both on a family of distributions from confident to flat; full derivations are in the supplementary material.

4 Where Measured Collapse Goes

Equation (2) splits a gradient; between two checkpoints a and b the change itself splits the same way. With $\mathbb{E}_x[S_m]$ the mean, over responses sampled from x, of model m's next-token entropies summed along the response,

$$\Delta H = \underbrace{\mathbb{E}_b[S_b] - \mathbb{E}_b[S_a]}_{L} + \underbrace{\mathbb{E}_b[S_a] - \mathbb{E}_a[S_a]}_{V}, \tag{3}$$

Table 2. Where the trajectory entropy lost between the base model ($H_a = 728.8$ nats per response) and a later checkpoint goes, in nats per response. L: same states, sharper conditionals; V: other states visited; $\Delta H = L + V$ exactly. Brackets: 95% prompt-bootstrap interval.

base →	ΔH	L	V	$V/\Delta H$
seed 1*, step* 100				
GRPO	−588.5	−58.1	−530.4	0.90 [0.89, 0.91]
PERMUTED	−627.3	−62.0	−565.3	0.90 [0.89, 0.91]
SIGNED	−616.0	−66.1	−550.0	0.89 [0.88, 0.90]
replication, other seeds				
GRPO, seed 2, step 80	−551.3	−38.2	−513.1	0.93 [0.92, 0.94]
SIGNED, seed 3, step 100	−613.8	−57.6	−556.2	0.91 [0.90, 0.92]

Table 3. Secondary intervals, seed 1, registered without a decision rule; nats per response. The SIGNED rows come from a 200-step re-run of that arm.

interval	ΔH	L	V	$V/\Delta H$
SIGNED 0 → 50	−590.4	−48.0	−542.4	0.92 [0.91, 0.93]
SIGNED 50 → 100	−25.6	−13.7	−11.9	0.46 [0.33, 0.55]
SIGNED 100 → 150	+9.0	+2.9	+6.1	0.67 [0.38, 0.78]
SIGNED 150 → 200	+40.6	+9.5	+31.1	0.77 [0.69, 0.82]
GRPO 80 → 110	+12.1	+2.9	+9.3	0.76 [0.65, 0.82]
PERMUTED 80 → 110	−15.9	−7.2	−8.6	0.54 [0.43, 0.62]

exactly. L is the change at the states b visits; V is the change in which states are visited, scored by one fixed model. The split depends on whose states are used, so we average it with the reverse ordering ($L' = \mathbb{E}_a[S_b] - \mathbb{E}_a[S_a]$, $V' = \mathbb{E}_b[S_b] - \mathbb{E}_a[S_b]$), which also adds to ΔH exactly.

Procedure. We use 512 training prompts of DAPO-Math-17k, the same for every checkpoint, and four responses per prompt sampled at temperature 1, top-p 1, up to 3072 tokens — the training sampler. Every response set of an interval is scored by both of its checkpoints with the full-vocabulary next-token entropy from an fp32 softmax. Intervals are 95% percentile bootstraps over prompts (10,000 resamples), paired across checkpoints. Two rules were registered before any sample was drawn: a measurable visitation share is claimed only if the interval of V excludes zero with the sign of ΔH; and a damping arm is called consistent with acting on the local channel only if its difference from GRPO in L excludes zero and exceeds its difference in V. A replication on other seeds was registered after the seed-1 result and before its samples; it uses the two retained checkpoints off seed 1 that match a primary arm.

Result. GRPO loses −588.5 nats per response, and the visitation channel carries 0.90 of it, which meets the registered criterion (Table 2, Fig. 2). The replication

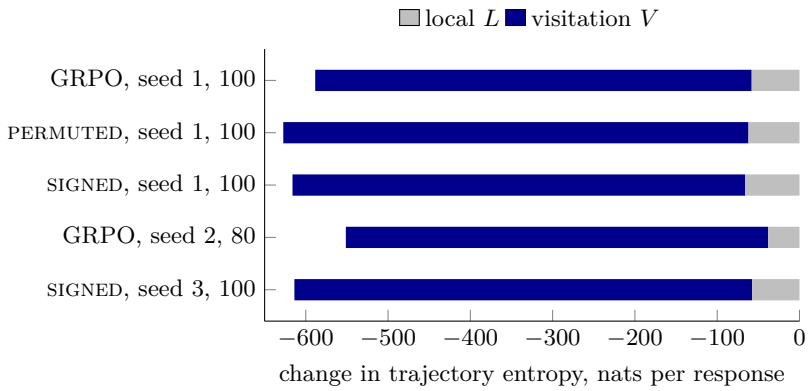


Fig. 2. The entropy lost from the base model to the given step, split into the two channels of Eq. (3) (values of Table 2). The local channel, the only one a token-level estimator sees, is the short grey segment in every run.

meets the same criterion: GRPO on seed 2 gives 0.93 and the campaign's SIGNED run on seed 3 gives 0.91. Most of the measured collapse is therefore not sharpening at the states the policy already visited: the trained policy moves to states where the base model was already confident, and the local channel, the only one a token-level estimator sees, carries the remainder. The two damping arms (Section 5.1) give nearly the same split as GRPO.

Timing. The loss is front-loaded (Table 3): by step 50 the SIGNED run has already lost −590.4 nats, 0.92 of it through visitation. Between later checkpoints the changes are much smaller and the local channel's part is larger (visitation share 0.46 from step 50 to 100), and where entropy rises late the rise again goes mostly through visitation.

Checks. Because $\mathcal{H}$ grows with response length and long responses are cut at 3072 tokens, we checked, after seeing the result, whether truncation or length carries the share: without truncated responses, per token, or both, it stays between 0.87 and 0.89 on seed 1 (Table 4) and between 0.88 and 0.92 on the replication. These variants were added after the result was seen and are exploratory. The second registered prediction is not borne out (Table 5): relative to GRPO the damping arms change V more than L, so the difference damping makes, integrated over training, is also mostly a difference in where the policy goes.

5 Why Aggregate Entropy Cannot See It

The visitation term as an advantage. Expanding (V) into a double sum over the position t of an entropy and the position u of a score, every term with $u \geq t$

Table 4. Exploratory variants of the visitation share, seed 1, base → step 100.

$V/\Delta H$	GRPO	PERM.	SIGNED
registered	0.90	0.90	0.89
no truncated	0.88	0.88	0.87
per token	0.89	0.89	0.88
both	0.88	0.88	0.87

Table 5. Each damping arm minus GRPO in the two channels, base → step 100, seed 1; nats per response, prompt-paired 95% intervals.

arm	ΔL	ΔV
PERM.	-3.9	-34.9
	$[-5.2, -2.7]$	$[-39.7, -30.3]$
SIGNED	-8.0	-19.6
	$[-9.4, -6.5]$	$[-24.1, -15.1]$

vanishes by causality, leaving a REINFORCE estimator whose reward-to-go is the remaining conditional entropy,

$$(\mathrm{V}) = \mathbb{E}\Big[\textstyle\sum_u \nabla_\theta \log \pi_\theta(y_u \mid s_u)\, H_{\mathrm{togo}}(s_u \oplus y_u)\Big], \tag{4}$$

with $H_{\mathrm{togo}}(s_u \oplus y_u) := \sum_{t>u} H(\pi_\theta(\cdot \mid s_t))$. Any function of s_u alone is an admissible baseline. We use the mean of H_{togo} over rollouts in the group whose prefixes $y_{<u}$ agree exactly, giving the advantage $A_H(s_u, y_u) := H_{\mathrm{togo}}(s_u \oplus y_u) - \bar{H}_{\mathrm{togo}}(s_u)$, non-zero only where siblings share a prefix and then diverge. Independent samples almost never share prefixes, so we schedule the sharing with a fixed tree (a root cut at depths $\{64, 192, 384\}$ with branching factor 2, $n = 8$ leaves), used only as a device for the estimator. With n leaves at most $n-1$ nodes branch, so at most $2(n-1)/T$ of positions carry $A_H \neq 0$ under any sampler.

The correction cannot move the mean. On each sibling set $\sum_j A_H(s_u, y_u^{(j)}) = 0$. A correction built on A_H can therefore change which sibling is damped but not how much damping there is on average, and an average over all response tokens should barely register it: aggregate token entropy is not a diagnostic for whether the visitation channel is acted on. The prediction it does make is that the spread *across siblings* widens.

From A_H to a token weight. H_{togo} involves states the update has not visited, so we forecast it from the hidden state with $\kappa = 2$ parallel multi-token-prediction heads [1], each calibrated by an affine map and summed with discount γ_H. Under the base method's assumption that one step moves only the sampled logit, the step changes the sampled token's log-probability by $\eta w_u(1 - \pi_u)$, with w_u the clipped advantage coefficient, which gives the visitation prediction $\mathrm{visit}_u = \eta w_u(1-\pi_u)\,\mathrm{clip}(A_H, \pm c)$. Because Ω is heavy-tailed, the visitation term is added *after* the base map, as a bounded adjustment on the support $\mathcal{S} = \{u : \mathrm{visit}_u \neq 0\}$,

$$\alpha_u^{\mathrm{f}} = \mathrm{clamp}\big(\alpha_u + \lambda\, b\, \tanh(\mathrm{visit}_u/\rho),\ \lambda_{\min}, 1\big), \tag{5}$$

with b the band width and ρ the RMS of visit over $\mathcal{S}$; at $\lambda = 0$ the pipeline is exactly the base method. Table 6 lists every approximation between the exact

Table 6. Every approximation between the exact decomposition and the implemented weight. A1–A4 are introduced by this work; B1–B4 are shared with the token-level method it extends. Derivations are in the supplementary material.

	approximation	what it discards	sign / bound	mitigation
—	Eqs. (2)–(4), A_H	nothing — identities	—	—
A1	horizon κ, discount γ_H	the tail beyond κ steps	under-estimate	common part cancels between siblings
A2	realized future $\rightarrow$ head marginal	mutual information with intervening tokens	over-estimate, monotone in k	A3
A3	per-head affine calibration	non-affine residual of A2	least squares	realized-entropy control arm
A4	clip A_H to $\pm c$	extreme branch scores	bounded by c	saturation logged
B1	first order in η	$O(\eta^2)$	—	small learning rate
B2	only the sampled logit moves	cross-position effects of shared weights	—	inherited
B3	rollouts from π_{old}	mismatch with π_θ	—	importance ratio
B4	self in the sibling set	$O(1/m)$ vs. leave-one-out	—	as in GRPO

identities and this weight. The correction is an instrument for testing the zero-sum prediction, not a proposed training method.

5.1 A test: aggregate entropy tracks the mechanism, not the signal

Setup. All runs fine-tune Qwen2.5-Math-1.5B [17] on DAPO-Math-17k [19] for 110 steps of 512 prompts with GRPO ($n = 8$, learning rate 10^{-6}, responses up to 3072 tokens, clip 0.2/0.28, no entropy bonus or KL). Five arms separate the *damping mechanism* (tree sampler, support, token-weight damping) from the *signal* A_H: GRPO; STEER (local Ω only); UNIFORM, which damps every supported position equally and so removes the signal while keeping its support; PERMUTED, which shuffles A_H within each sibling set, keeping magnitudes and the zero sum but destroying the branch–score assignment; and SIGNED, the correction as derived (Table 7). Every run validates on AIME24 (30 problems, 32 samples) every 10 steps; one evaluation has an across-problem standard error of .0282. Each run is reduced to its mean over steps 40–110, arms are paired within a seed, and contrasts use a paired t over the 3 seeds on which all arms finished $(1, 3, 4)$. The plan was registered before the runs.

Result. Across all three paired seeds, changing the damping mechanism shifts aggregate entropy far more than changing the signal with the mechanism held

Table 7. The five arms. UNIFORM and PERMUTED keep the damping mechanism and remove the branch-specific A_H signal in two different ways.

arm	rollout	token weight	signal in the weight
GRPO	plain	none	none
STEER	plain	local Ω	none
UNIFORM	tree	Ω + damping	support only
PERMUTED	tree	Ω + damping	A_H shuffled within siblings
SIGNED	tree	Ω + damping	A_H as derived

Table 8. Converged token-mean entropy, trajectory entropy $\mathcal{H}$ (nats per response) and AIME24 acc@32, paired within a seed (df $= n - 1$), grouped by what each contrast changes.

contrast	n	Δent.	t	$\Delta\mathcal{H}$	t	Δacc	t
the damping mechanism changes, the A_H signal does not							
STEER − GRPO	3	−.0419	−5.87	−42.5	−4.47	−.0065	−1.24
UNIF − STEER	3	+.0184	+13.09	+16.6	+4.43	+.0051	+1.37
the A_H signal changes, the damping mechanism does not							
SIGNED − UNIF	3	−.0077	−1.42	−6.4	−1.39	+.0069	+1.16
SIGNED − PERM	3	+.0044	+2.15	+1.8	+0.99	+.0087	+3.11

fixed (Table 8): turning the damping on shifts it by −.0419 and adding the tree with uniform damping by +.0184, whereas removing the signal shifts it by −.0077 and destroying the branch–score assignment by +.0044. The split survives length correction: on $\mathcal{H}$ the mechanism contrasts are −42.5 and +16.6 nats per response and the signal contrasts −6.4 and +1.8. Aggregate entropy reflects the damping mechanism, not the visitation signal, which is the dissociation the zero sum predicts. Three seeds establish the pattern more firmly than its effect sizes; the positive prediction, that the spread across siblings widens, is not measured.

Accuracy. The registered accuracy test, SIGNED − UNIFORM, straddles zero across seeds (−.0020 to +.0182), and the gaps between arms are smaller than GRPO's own spread across seeds, .0210, so we make no accuracy claim. Per-arm results, ablations and the full registration are in the supplementary material.

6 Discussion

What the split says about entropy control. Token-level methods damp the update at the sampled token in proportion to how much it sharpens that state. The measured split says that, over the first hundred steps, this is the smaller part of what is lost. The larger part is that the policy concentrates on prefixes along which the base model's own conditionals were already sharp. This is consistent with the observation that answer-level diversity can fall while token-level entropy stays

high [16]: per-state entropies say little about how concentrated the trajectory distribution has become, and an intervention that protects them leaves the larger part untouched.

Why damping moves V. Damping is applied token by token, yet its integrated effect relative to GRPO was larger in V than in L (Table 5). One reading is that a token weight changes which updates are made, and the trajectory distribution compounds those changes along the response, so a local intervention can act mostly through visitation over training. We did not test this mechanism, and the registered prediction that damping acts on the local channel is reported as not borne out.

A diagnostic for LLM training runs. Read as data, a training run is a sequence of checkpoints, and the split turns any two of them into an attribution of the entropy lost between them. It needs only responses sampled from both checkpoints and a forward pass of each model over both response sets — no gradients and no access to the training code — so it applies to logged and released checkpoints alike, and it can be tracked alongside reward and loss in place of the mean token entropy that cannot register the larger channel. With prompt-paired intervals its share is well resolved at a few hundred prompts (Table 2).

What it does not show. The split is descriptive: it says where entropy went, not whether keeping it would have helped, and accuracy did not respond to the visitation signal in our arms (Section 5.1). Whether path-level objectives slow the visitation part of collapse, and whether that improves reasoning, is what this measurement makes testable.

7 Limitations and Conclusion

Limitations. Every result is on one 1.5B mathematics model. The decomposition rests on seed 1 and one retained checkpoint each of seeds 2 and 3, so the seed-2 point is at step 80; it measures the split between checkpoints, not the gradient's, and the seed-1 SIGNED checkpoint is a re-run of that configuration on different hardware. The dissociation uses three seeds and AIME24's 30 problems for validation. The local–visitation trade-off at real branch points is derived, not measured, and the A_H estimator rests on an approximation inherited from STEER (an update changes only the sampled logit) that is false for shared parameters.

Conclusion. Entropy collapse in RLVR is mostly a change in where the policy goes. Measured between checkpoints, 0.90 of the trajectory entropy GRPO loses goes through the visitation channel, a share that replicates on two further seeds, and token-level entropy control acts only on the remainder. The local estimator vanishes, for a correct reason, exactly where the visitation term is largest; the visitation term is estimable but sums to zero over siblings, so aggregate entropy

cannot register it, and the experiment observes that dissociation. Whether the share holds across model scales, and which path-level objectives preserve diversity best, are open.

Disclosure of AI use. An AI assistant was used for writing code, running and checking analyses, literature search, and drafting and editing the text; the authors evaluated its suggestions, reviewed the code, checked every claim and citation, and take full responsibility for the content. Every number is emitted by a script from the training and evaluation logs.

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